

Using the selected papers, synthesize the literature on digital prosthetic customization platforms with a focus on two related questions: first, how these systems structure the experience of technical professionals such as designers, prosthetists, clinicians, and technicians, in comparison with the experience of end users; and second, how these platforms mediate the design process through parametric configuration, visualization, supervision, and human-in-the-loop assistance. Distinguish clearly between direct evidence from upper-limb prosthetic customization, transferable evidence from adjacent domains such as lower-limb prosthetics, orthotics, rehabilitation devices, and assistive technologies, and explicit literature gaps. Focus especially on trust, intelligibility, participation in configuration, perceived control, supervision, iterative adjustment, variant comparison, decision-making responsibility, differentiated user roles, and the platform’s function as a space of asymmetric collaboration rather than a purely neutral interface.

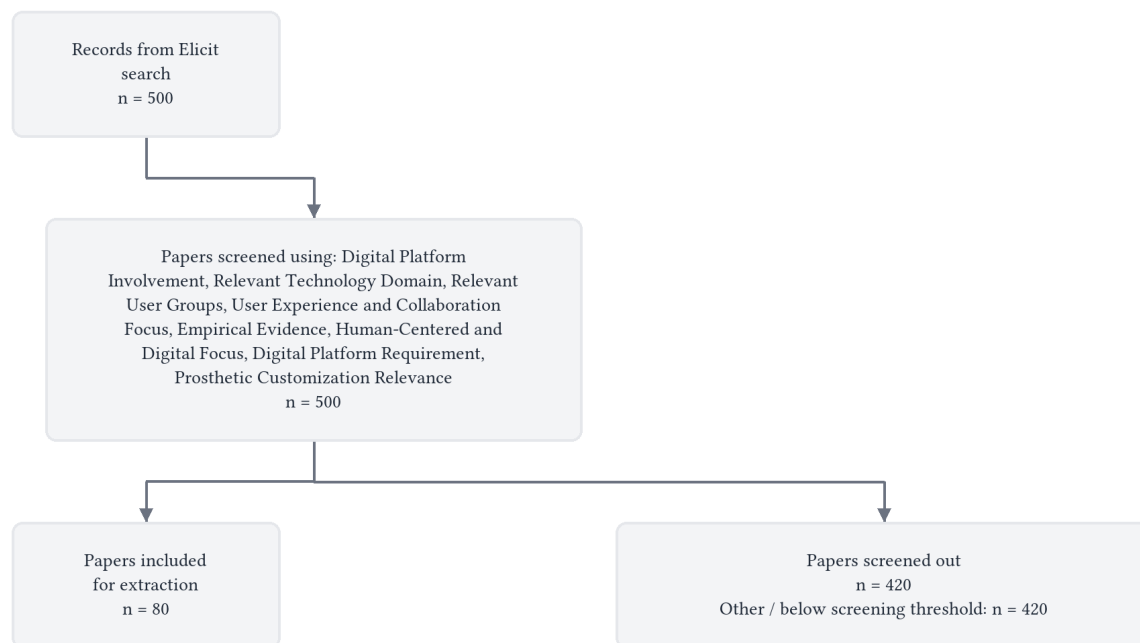
Digital prosthetic customisation platforms operate as asymmetric collaboration spaces that concentrate configurative authority and decision-making responsibility with technical professionals in socket design contexts while granting end users meaningful but supervised agency in control customisation contexts, with automation functioning throughout as a cognitive offload mechanism that encodes clinical expertise rather than replacing it—and the field’s central gap is the near-absence of direct evidence for upper-limb socket design platforms, leaving the transferability of well-developed lower-limb paradigms unvalidated.

Abstract

Digital prosthetic customisation platforms consistently function as asymmetric collaboration spaces rather than neutral interfaces, structuring differentiated experiences for technical professionals and end users in ways that reflect and reinforce existing clinical hierarchies. In socket design systems—where the evidence base is dominated by lower-limb transferable work—prosthetists and orthopaedic technicians are the primary configurative agents, with platforms encoding their tacit knowledge as automated rules and virtual tools [1, 2], while patients serve primarily as data sources and comfort evaluators rather than design participants [3, 4]. Parametric automation, FEA-based simulation, and evidence-generated design approaches can match clinician-led designs on aggregate comfort outcomes—the Mbithi et al. crossover trials found no statistically significant difference between evidence-generated and CAD/CAM sockets (median SCS 8.6 vs. 8.8, $p = 0.43$) [5]—but qualitative findings consistently show that clinical expertise retains irreplaceable relational, communicative, and contextually adaptive functions that algorithms do not reproduce [5, 6]. In control customisation platforms, by contrast, end users hold meaningful configurative authority: systems for user-preferred tuning of robotic knees [7], semi-autonomous gaze-EMG interfaces [8], and multimodal shared control [9] all demonstrate that user preference over control behaviour is something only the user can determine, and that automation functions most effectively as a cognitive offload mechanism that amplifies rather than displaces user agency. Intelligibility of platform logic emerges as a distinct dimension of user experience: making classifier decision boundaries directly visible to users improved myoelectric control proficiency [10], and training strategies oriented

toward class separability produced more accurate user mental models of system behaviour [11], while clinicians performing digital socket rectification reported reduced confidence due to the loss of tactile feedback that traditional casting provides [6]. Supervision structures are well developed in socket design contexts through FEA pressure visualisation and tiered parameter access [12, 13], but less formalised in control systems. The most consequential gap in the literature concerns upper-limb socket design specifically: despite substantial evidence from lower-limb digital fabrication platforms, directly analogous systems for trans-radial and trans-humeral sockets remain underdeveloped and undertested [14, 15], and whether lower-limb design paradigms, parametric tools, and evidence-generation methods transfer to upper-limb biomechanical and functional requirements is an empirical question the current corpus cannot answer.

Flow Diagram



Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "Using the selected papers, synthesize the literature on digital prosthetic customization platforms with a focus on two related questions: first, how these systems structure the experience of technical professionals such as designers, prosthetists, clinicians, and technicians, in comparison with the experience of end users; and second, how these platforms mediate the design process through parametric configuration, visualization, supervision, and human-in-the-loop assistance. Distinguish clearly between direct evidence from upper-limb prosthetic

customization, transferable evidence from adjacent domains such as lower-limb prosthetics, orthotics, rehabilitation devices, and assistive technologies, and explicit literature gaps. Focus especially on trust, intelligibility, participation in configuration, perceived control, supervision, iterative adjustment, variant comparison, decision-making responsibility, differentiated user roles, and the platform’s function as a space of asymmetric collaboration rather than a purely neutral interface. ”

The search returned 500 total results from Elicit.

We retrieved 500 papers most relevant to the query for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Digital Platform Involvement:** Does the study involve digital platforms, software systems, or computer-aided tools for prosthetic customization, design, or configuration?
- **Relevant Technology Domain:** Does the study examine upper-limb prosthetics, lower-limb prosthetics, orthotics, rehabilitation devices, or assistive technologies in the context of digital customization?
- **Relevant User Groups:** Does the study involve any of the following user groups: prosthetists, clinicians, designers, technicians, engineers, or end users/patients?
- **User Experience and Collaboration Focus:** Does the study examine user experience, human-computer interaction, user interface design, collaboration, decision-making processes, user roles, trust, control, or supervision in prosthetic/assistive technology contexts?
- **Empirical Evidence:** Does the study provide empirical data or substantial literature synthesis (excluding conference abstracts, editorials, opinion pieces, or commentaries without substantive evidence)?
- **Human-Centered and Digital Focus:** Does the study consider user experience or digital platforms rather than focusing solely on mechanical design, materials science, or biomechanical performance without human factors?
- **Digital Platform Requirement:** Does the study involve digital platform components rather than examining only traditional (non-digital) prosthetic design, fitting, or manufacturing processes?
- **Prosthetic Customization Relevance:** Does the study address prosthetic customization platforms rather than focusing exclusively on surgical procedures, medical treatments, clinical outcomes, cost-effectiveness, health-care policy, or insurance coverage without reference to customization platforms?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

At abstract screening, the number of papers excluded for each primary reason was:

- **Other / below screening threshold:** n = 420

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Platform Type:**

Extract details about the digital prosthetic customization platform or system, including:

- Type of platform (design software, VR training system, CAD/CAM system, control interface, etc.)
- Specific customization capabilities offered

- Whether it's for upper-limb prosthetics (direct evidence), adjacent domains like lower-limb/orthotics/assistive tech (transferable evidence), or other
- Level of automation vs human control in the customization process

- **User Roles:**

Extract information about different user types and their distinct roles in the platform, including:

- Technical professionals involved (designers, prosthetists, clinicians, technicians, researchers)
- End users (prosthesis users, patients, trainees)
- How each role's experience with the platform differs
- Specific tasks or responsibilities assigned to each user type
- Any role-based access controls or interface variations

- **Collaboration Dynamics:**

Extract evidence of how the platform mediates collaboration between different users, focusing on:

- Whether collaboration is symmetric (equal roles) or asymmetric (differentiated roles)
- How technical professionals and end users interact through the platform
- Co-design processes, shared decision-making, or hierarchical workflows
- Communication mechanisms built into the platform
- Evidence of the platform functioning as neutral interface vs. structured collaboration space

- **Trust and Control:**

Extract findings related to trust, perceived control, and decision-making responsibility, including:

- User trust in platform recommendations or automated features
- Perceived control over customization decisions by different user types
- Who has final decision-making authority in the design process
- Intelligibility of platform algorithms or suggestions to users
- User confidence in platform-generated vs human-generated designs

- **Configuration Mechanisms:**

Extract details about how the platform enables parametric configuration and design mediation, including:

- Parametric design tools and configuration options available
- Visualization capabilities (3D modeling, simulation, AR/VR, etc.)
- Iterative adjustment processes and variant comparison features
- Human-in-the-loop assistance mechanisms
- Supervision tools for oversight of design decisions

- **User Experience:**

Extract findings about user experience across different user types, focusing on:

- Usability assessments, workload measures, or satisfaction scores
- Differences in experience between technical professionals and end users
- Participation levels in configuration processes
- Training requirements or learning curves for different user types
- Preferences for platform-mediated vs traditional design processes

- **Design Process Impact:**

Extract evidence of how the platform changes or structures the prosthetic design process, including:

- Comparison with traditional design workflows
- Time efficiency, iteration cycles, or design quality impacts
- How the platform influences professional expertise application
- Changes in patient/user involvement in design decisions
- Integration challenges with existing clinical or manufacturing workflows

- **Key Findings:**

Extract the main findings relevant to platform-mediated prosthetic customization, including:

- Primary outcomes related to trust, intelligibility, control, or collaboration
- Evidence of platform effectiveness for different user types
- Identified benefits, limitations, or barriers to platform use
- Recommendations for platform design or implementation
- Any unexpected or counterintuitive results about user roles or collaboration

- **Evidence Quality:**

Extract information about the strength and transferability of evidence, including:

- Study design and sample characteristics
- Whether findings are from direct upper-limb prosthetic studies or adjacent domains
- Explicitly identified literature gaps mentioned by authors
- Limitations affecting generalizability to prosthetic customization contexts
- Authors' assessments of evidence quality or research needs

Results

Characteristics of Included Studies

The 80 sources span publications from 2002 to 2026 and encompass a broad range of platform types, application domains, and study designs. The table below characterizes each source according to its platform type, primary domain, and study focus relevant to the research questions.

Study	Full Text Retrieved?	Platform Type	Primary Domain	Study Focus
A. Vitali, 2016 [16]	No (abstract only)	CAD/CAM system (SMA2)	Lower-limb prosthetics (transferable)	Socket design; NUI; additive manufacturing
Stella Gabbiadini, 2011 [1]	No (abstract only)	Knowledge-based design framework	Lower-limb prosthetics (transferable)	Knowledge-guided socket and prosthesis design
Sébastien Mick et al., 2024 [17]	Yes	Control interface / robotic arm platform	Upper-limb prosthetics (direct)	Customizable trans-humeral prosthesis test-bed

Study	Full Text Retrieved?	Platform Type	Primary Domain	Study Focus
G. Colombo et al., 2010 [3]	No (abstract only)	Integrated 3D design paradigm	Lower-limb prosthetics (transferable)	Reverse engineering, simulation, rapid prototyping
L. Resnik et al., 2018 [18]	Yes	EMG pattern recognition control interface	Upper-limb prosthetics (direct)	User experience of EMG-PR with DEKA Arm
F. Mbithi et al., 2025 [5]	Yes	CAD/CAM (evidence-generated sockets)	Lower-limb prosthetics (transferable)	Comfort of data-driven vs. clinician-led sockets
Michelle Esponda & Thomas Howard, 2018 [19]	No (abstract only)	Adaptive multi-modal control interface	Upper-limb prosthetics (direct)	Adaptive grasp control via multi-modal sensing
Trevor Binedell et al., 2020 [20]	No (abstract only)	CAD/CAM + 3D printing	Upper-limb prosthetics (direct)	Custom 3D-printed forequarter prosthesis
G. Colombo et al., 2010a [21]	No (abstract only)	Design software / CAD system	Lower-limb prosthetics (transferable)	Virtual configuration framework for prosthesis
Julio Fajardo et al., 2017 [22]	No (abstract only)	Open-source CAD/CAM + control interface	Upper-limb prosthetics (direct)	Affordable open-source myoelectric hand
Jennifer Olsen et al., 2020 [14]	Yes	CAD/CAM + optical scanning + 3D printing	Upper-limb prosthetics (direct)	Digital trans-radial socket manufacturing
F. Mbithi et al., 2024 [4]	Yes	CAD/CAM (Tracer software)	Lower-limb prosthetics (transferable)	Evidence-generated vs. conventional socket design
G. Colombo et al., 2011 [23]	No (abstract only)	Design software for prosthesis configuration	Lower-limb prosthetics (transferable)	Knowledge-guided design framework validation
F. Górski et al., 2023 [24]	Yes	XR (VR/AR/MR) + CAD/CAM (AutoMedPrint)	Upper-limb prosthetics (direct)	Extended reality in prosthesis development
A. Mayo et al., 2022 [25]	No (abstract only)	CAD/CAM + 3D printing	Lower-limb prosthetics (transferable)	Stakeholder perceptions of digital socket fabrication
Duncan R. C. Lee et al., 2024 [26]	Yes	Physics-based digital design framework	Lower-limb prosthetics (transferable)	FEA-based transtibial liner-socket design
D. Regazzoni et al., 2019 [27]	No (abstract only)	Virtual platform (CAD + FEM + MOCAP)	Lower-limb prosthetics (transferable)	Integrated virtual design and assessment platform

Study	Full Text Retrieved?	Platform Type	Primary Domain	Study Focus
Clara Phillips et al., 2023 [28]	No (abstract only)	Digital technology (3D scanning + printing)	Lower-limb prosthetics (transferable)	Patient perceptions of digital socket manufacturing
Sarah Whatley et al., 2023 [29]	Yes	Generative design algorithm	Lower-limb prosthetics (transferable)	Aesthetic personalisation via motion capture and algorithms
Nienke Kerver et al., 2023 [30]	No (abstract only)	Patient decision aid (digital)	Upper-limb prosthetics (direct)	Decision support for terminal device selection
G. Colombo et al., 2013 [31]	No (abstract only)	Virtual environment (full prosthesis design)	Lower-limb prosthetics (transferable)	Full virtual design and gait simulation platform
Lee-Kuen Chua et al., 2014 [32]	No (abstract only)	Haptic body-powered prosthesis simulator	Upper-limb prosthetics (direct)	Parameter optimisation simulator
G. Colombo et al., 2015 [33]	Yes	NUI / CAD system (Leap Motion)	Lower-limb prosthetics (transferable)	Hand-tracking devices for socket design
Briana N. Perry et al., 2018 [34]	Yes	VR training system	Upper-limb prosthetics (direct)	VR-based myoelectric prosthesis training
Feng Zhou et al., 2023 [35]	No (abstract only)	Generative co-design algorithm	Lower-limb prosthetics (transferable)	Aesthetic prosthetic co-design with algorithms
Daniel Panchik et al., 2021 [36]	Yes	CAD/CAM (OpenSCAD) + 3D printing	Upper-limb prosthetics (direct)	Task-specific digit prosthesis design
G. Colombo et al., 2014 [37]	No (abstract only)	Design software + hand-tracking	Lower-limb prosthetics (transferable)	Low-cost hand-tracking for socket design
Abbas Alili et al., 2023 [7]	Yes	User-controlled interface + RL algorithm	Lower-limb prosthetics (transferable)	User-preferred tuning for robotic knee
Ming-Huang Lin, 2023 [38]	No (abstract only)	VR collaborative design system	Upper-limb prosthetics (direct)	VR co-creation of prosthetic accessories
Wei Li et al., 2025 [39]	No (abstract only)	ML + AR control interface	Upper-limb prosthetics (direct)	Closed-loop co-adaptation for myoelectric control
Rebecca J. Greene et al., 2024 [8]	No (abstract only)	Semi-autonomous hybrid gaze-EMG interface	Upper-limb prosthetics (direct)	SA interface for multi-DOF prosthesis control

Study	Full Text Retrieved?	Platform Type	Primary Domain	Study Focus
Sara Peixoto et al., 2025 [40]	Yes	Online personalisation platform	Upper-limb prosthetics (direct)	Aesthetic personalisation interface design
A. Vitali et al., 2019 [41]	No (abstract only)	Virtual platform (reverse engineering + MOCAP)	Lower-limb prosthetics (transferable)	Virtual platform for design and gait assessment
Chunhao Peng et al., 2025 [9]	No (abstract only)	Multimodal shared control interface	Upper-limb prosthetics (direct)	Multimodal fusion for prosthetic hand/wrist
InHwa Lee et al., 2026 [42]	No (abstract only)	XR myoelectric training system (MyoTrainXR)	Upper-limb prosthetics (direct)	User-centered XR myoelectric training
G. Colombo et al., 2018 [2]	Yes	CAD system (SMA ²)	Lower-limb prosthetics (transferable)	Interactive 3D socket design environment
Vishal Pendse et al., 2026 [43]	No (abstract only)	Predictive algorithm / CAD (PACT)	Upper-limb prosthetics (direct)	Automated transradial socket design algorithm
Junghun Lee et al., 2023 [44]	No (abstract only)	Computer vision + automated socket generation	Upper-limb prosthetics (direct)	Low-barrier 3D prosthetic design via mobile phone
D. Bonacini & Gerardina Buldo, 2024 [45]	Yes	CAD/CAM + FEM + additive manufacturing	Lower-limb prosthetics (transferable)	Digital methodology for BK socket manufacture
N. Patiniott et al., 2025 [46]	Yes	Ontology-driven cyber-physical service system	Lower-limb prosthetics (transferable)	Adaptive personalised prosthesis aftercare
David Quintero et al., 2018 [47]	No (abstract only)	Clinician control interface (CCI)	Lower-limb prosthetics (transferable)	Intuitive clinical tuning of powered knee-ankle
Artin Saberpour Abadian et al., 2023 [48]	No (abstract only)	Computational design tool (WRLKit)	Upper-limb / assistive technology (adjacent)	Personalised wearable robotic limb design
Vaynee Sungeelee et al., 2024 [11]	Yes	ML-based prosthesis control training interface	Upper-limb prosthetics (direct)	Teaching strategies for ML-based myoelectric arm
A. Arunachalam et al., 2019 [49]	No (abstract only)	Hierarchical optimisation control interface	Upper-limb prosthetics (direct)	Cost-function-based prosthesis tuning
A. Eshraghi et al., 2024 [6]	Yes	CAD/CAM (OMEGA software)	Lower-limb prosthetics (transferable)	Manual vs. digital socket design feasibility
Mixuan Li & L. Aflatoony, 2025 [50]	Yes	Parametric design tool (Grasshopper-based)	Assistive technology (adjacent)	OT-facing parametric tool for hand grips

Study	Full Text Retrieved?	Platform Type	Primary Domain	Study Focus
Enzo Romero et al., 2025 [51]	No (abstract only)	CAD/CAM + AI control interface	Upper-limb prosthetics (direct)	Affordable AI-driven myoelectric prosthesis
Heather E. Williams et al., 2023 [52]	Yes	Simulated prosthesis platform (MAPP)	Upper-limb prosthetics (direct)	Evaluation metrics for myoelectric controllers
Andru Liu et al., 2023 [53]	No (abstract only)	VR games + adaptive LDA model	Upper-limb prosthetics (direct)	Semi-supervised adaptation via virtual gameplay
J. Wojtusich et al., 2012 [54]	No (abstract only)	Biomechanical modelling and simulation	Lower-limb prosthetics (transferable)	User-specific biomechanical simulation environment
Maria Vlachostergiou, 2022 [55]	No (abstract only)	AR visualisation + co-design process	Upper-limb prosthetics (direct)	Co-creative speculative prosthetic design
Julio Fajardo et al., 2018 [56]	No (abstract only)	Hybrid UPI (object classification + EMG)	Upper-limb prosthetics (direct)	Object-classification-based prosthesis control
G. Colombo et al., 2013a [57]	No (abstract only)	CAD/CAM + Leap Motion + haptic devices	Lower-limb prosthetics (transferable)	Socket virtual design with low-cost devices
Ricardo Garcia-Rosas et al., 2020 [58]	No (abstract only)	Data-driven personalisation control interface	Upper-limb prosthetics (direct)	Implicit motor behaviour for prosthetic personalisation
D. Hepp et al., 2015 [59]	Yes	Smartphone-based control interface	Upper-limb prosthetics (direct)	Smartphone-supported upper limb prosthesis
J. Howard et al., 2024 [60]	No (abstract only)	CAD/CAM co-design framework	Assistive technology (adjacent)	Barriers to scaling co-design AT provision
Gianmarco Cirelli et al., 2026 [61]	No (abstract only)	Semi-autonomous EMG + vision control	Upper-limb prosthetics (direct)	Context-aware semiautonomous prosthetic control
Giuseppe Stracquadanio et al., 2025 [62]	No (abstract only)	Eye-in-hand grasp system (shared autonomy)	Upper-limb prosthetics (direct)	Shared-autonomy prosthetic grasping
Donato Brusamento et al., 2020 [63]	Yes	Closed-loop myocontrol training interface	Upper-limb prosthetics (direct)	Closed-loop data acquisition for myocontrol
Dimitrios Menychtas et al., 2019 [64]	No (abstract only)	Simulation algorithm + GUI	Upper-limb prosthetics (direct)	Motion simulation for prosthesis prescription

Study	Full Text Retrieved?	Platform Type	Primary Domain	Study Focus
Ricardo Garcia-Rosas et al., 2026 [65]	No (abstract only)	Commercial VR training system (Myo-Hand XP)	Upper-limb prosthetics (direct)	Co-designed VR myoelectric training, commercial
Bretta L. Fylstra et al., 2022 [66]	Yes	RL auto-tuning + visual feedback	Lower-limb prosthetics (transferable)	Human-prosthesis cooperation in gait tuning
D. Wells et al., 2002 [67]	No (abstract only)	Clinician software tools (MyoWizard/MyoAssistant)	Upper-limb prosthetics (direct)	Software tools for clinician control customisation
Emma Reznick et al., 2024 [68]	No (abstract only)	Clinical tuning interface (CTI)	Lower-limb prosthetics (transferable)	Clinical tuning interface for powered knee-ankle
Ruichen Yang et al., 2025 [10]	Yes	3D visual feedback interface (Reviewer)	Upper-limb prosthetics (direct)	Visual pattern separability feedback for PR control
Daniel C. Norvell et al., 2025 [69]	No (abstract only)	Web-based clinical decision support tool	Lower-limb prosthetics (transferable)	Evidence-based prosthetic mobility prediction tool
Ai Reen Chu et al., 2024 [70]	No (abstract only)	Participatory design platform	Upper-limb prosthetics (direct)	Collaborative 3D-printed prosthesis design protocol
Fahd Ahmed et al., 2025 [71]	No (abstract only)	AR-enhanced design and fitting system	Prosthetics general (direct)	AR + ML for real-time prosthetic customisation
D. Regazzoni et al., 2016 [72]	No (abstract only)	DHM-based virtual gait analysis tool	Lower-limb prosthetics (transferable)	Gait and pressure visualisation platform
Huimin Hu et al., 2025 [73]	Yes	Closed-loop VR rehabilitation platform	Upper-limb prosthetics (direct)	Closed-loop assessment and VR training
R. Morotti, 2014 [12]	No (abstract only)	Virtual testing lab (FEA + CAD integration)	Lower-limb prosthetics (transferable)	Automated FEA simulation for socket optimisation
Marko Marković, 2016 [74]	No (abstract only)	Sensor-fusion control interface	Upper-limb prosthetics (direct)	Sensor-fusion and feedback for myoelectric control
S. Rozevink et al., 2025 [75]	Yes	VR training environment	Upper-limb prosthetics (direct)	VR prosthesis control training usability
Xiaolu Bai et al., 2024 [13]	Yes	User-controlled interface (UCI) framework	Lower-limb prosthetics (transferable)	Human factors in intelligent LLP interaction

Study	Full Text Retrieved?	Platform Type	Primary Domain	Study Focus
Yunmei Liu et al., 2026 [76]	No (abstract only)	VR platform (prosthetic control evaluation)	Upper-limb prosthetics (direct)	VR for early-stage prosthetic control evaluation
Luca Gabriele De Vivo Nicoloso et al., 2021 [77]	Yes	CAD/CAM + 3D printing (monocoque design)	Lower-limb prosthetics (transferable)	Bio-inspired 3D-printed transtibial prosthesis
Alessandro Visconti et al., 2024 [78]	No (abstract only)	VR with eye-tracking + vibrotactile feedback	Upper-limb prosthetics (direct)	VR embodiment enhancement for bionic limb
T. Bodkin, 2017 [79]	No (abstract only)	Hybrid CAD specification study	Lower-limb prosthetics (transferable)	CAD system specification for prosthetic socket design
J. Vézien, 2007 [80]	No (abstract only)	CAD/CAM + virtual environment	Rehabilitation/assistive devices (adjacent)	User-customised design of rehabilitation aids
B. Oldfrey et al., 2024 [15]	Yes	CAD/CAM + additive manufacturing (scoping review)	Prosthetics (primarily upper-limb)	Scoping review of digital fabrication in prosthetics

The corpus divides broadly into three platform categories: physical design and socket fabrication systems (predominantly lower-limb, with growing upper-limb representation), control customisation interfaces (predominantly upper-limb), and training and visualisation environments (primarily upper-limb). Of the 80 sources, approximately 42 provide direct evidence from upper-limb prosthetic contexts, while the remaining 38 offer transferable evidence from lower-limb prosthetics, orthotics, rehabilitation devices, or assistive technology. Full texts were retrieved for 24 sources; the remaining 56 were available only as abstracts, which constrains the depth of extraction for those entries and limits conclusions about internal study details.

Differentiated User Roles and Structured Experiences

The Asymmetric Architecture of Professional vs. End-User Access

Across the corpus, the most consistent structural feature of digital prosthetic customisation platforms is their asymmetric organisation of user roles. Rather than providing a single interface to all participants, platforms consistently partition access, capability, and responsibility between technical professionals and end users. This asymmetry is not incidental but constitutes the foundational logic of how these systems are designed.

On the technical professional side, the depth and type of configuration access varies by discipline. Prosthetists and orthopaedic technicians are the primary operators of socket design platforms. In the Socket Modelling Assistant (SMA/SMA²) series developed at the University of Bergamo, the tool is explicitly oriented around the workflow and

tacit knowledge of the orthopaedic technician, encoding their traditional manipulations as virtual operations [1, 2]. SMA² provides interactive tools including automatic rectification, mesh editing, and template-based trim-line generation, all structured to reflect how technicians work rather than how patients experience the device [2]. The platform is explicitly described as making design “more reliable and repeatable compared to the conventional process, but still enough similar to it to be accepted by the involved personnel” [2], underscoring that professional acceptance, not patient-centricity, anchors the design rationale. Similarly, the knowledge-based framework described by Gabbiadini integrates expert knowledge via virtual assistants that guide technicians step by step through selection and sizing decisions, with automated execution “when possible” [1]. In this architecture, the platform amplifies professional expertise rather than distributing it.

Clinicians occupy a distinct but also privileged role in control customisation contexts. The early MyoWizard and MyoAssistant tools described by Wells et al. provided prosthetists and therapists with the ability to select pre-defined control strategies, program them into micro-controllers, and fine-tune gain and threshold parameters through a graphical signal display [67, 67]. This clinician-centric model persists through the 2024 Clinical Tuning Interface (CTI) for powered knee-ankle prostheses, which translates clinical tuning concepts directly into model adjustments, enabling a prosthetist to complete full multi-activity configuration in under 20 minutes compared with hours of engineer-led parameter adjustment [68, 68]. The CTI concentrates final decision-making authority in the prosthetist [68]. Quintero et al.'s clinician control interface for a powered knee-ankle prosthesis likewise explicitly repositions the prosthetist as the operator of a virtual constraint scheme, replacing the researcher or engineer as the primary tuner [47, 47]. Both systems function as structured collaboration spaces in which clinical authority is formalised through interface design.

End users, by contrast, are most commonly positioned as subjects of measurement, providers of subjective feedback, or beneficiaries of decisions made upstream. In the majority of socket design platforms—including the Colombo et al. paradigm series [3, 21, 23], the Morotti virtual testing lab [12], and the Bonacini and Buldo digital methodology [45]—patients provide data (residual limb scans, gait recordings, pressure measurements) and receive devices, but are not described as operating design interfaces or making configuration selections. The patient's role as a data source rather than a design agent is a consistent structural characteristic of socket design platforms, regardless of whether the domain is lower- or upper-limb. Even in studies where patient comfort scores are the primary outcome measure, such as the Mbithi et al. crossover trials of evidence-generated sockets [4, 5], the patient's function is evaluative rather than configurative: they wear both socket types and report comfort, but the design decisions are made by prosthetists using data from previous clinical records.

There are meaningful exceptions and gradations. In prosthesis control platforms, end users are considerably more active as co-configurers. The robotic knee tuning framework of Alili et al. provides a User-Controlled Interface that allows users to select their preferred knee kinematic profiles by modifying control points in a four-dimensional space [7, 7], with a reinforcement learning algorithm then adjusting the 12 underlying impedance parameters to match those preferences. Users in that study could accurately and consistently identify their preferred control profiles in blinded comparative testing [7], suggesting genuine end-user participation in functional configuration rather than merely aesthetic or comfort feedback. The Chunhao Peng et al. multimodal shared control system similarly frames users as having “full control authority” via EMG and voice, with system autonomy acting as a facilitator rather than a replacement of user intent [9]. The smartphone-based control platform of Hepp et al. lets patients preselect and switch grip patterns and choose from three control algorithms, placing meaningful functional customisation in the hands of the user [59, 59].

The Peixoto et al. online personalisation platform for myoelectric prostheses represents the furthest shift toward end-user autonomy in the corpus: it is explicitly designed to allow people with upper limb amputation to personalise aesthetics independently, with digital accessibility features, a support chatbot, and confirmatory interaction

mechanisms [40, 40, 40]. System Usability Scale scores of 79.5 to 92.5 across three participant groups [40] indicate that the inclusive design approach succeeded in making the platform genuinely accessible to end users without professional mediation. However, this platform focuses on aesthetic rather than functional configuration, which limits its comparability with socket design or control tuning systems.

Role Differentiation by Platform Type

The pattern across the corpus is that the degree of end-user participation in configuration is strongly determined by platform type. In socket fabrication and physical design systems, end users have little configurative agency; in control customisation interfaces, they have moderate to high agency; in training and visualisation platforms, they are the primary operators, though under technical professional supervision.

This pattern is illustrated by contrasting the Regazzoni et al. virtual platform for lower-limb prosthesis design [27]—in which prosthetists and technicians are the primary users and the application is noted as useful for “training purpose for junior orthopedic technicians” [27]—with the VR training system of Perry et al. [34], in which individuals with upper extremity loss are the primary operators of a stereoscopic, motion-controlled avatar environment, following and controlling virtual limb representations as a training intervention. In the latter, technical professionals develop and administer the platform but are not its operational users [34]. The Rozevink et al. study of a VR prosthesis control training environment similarly found that both therapists and prosthesis users operated the system, but with meaningfully different profiles: therapists scored significantly higher on subscales of conscious processing and visual attention [75], reflecting their professional surveillance function, while prosthesis users experienced lower perceived competence despite similar enjoyment and usability scores.

Occupational therapists constitute a third distinct user category in the corpus, sitting between prosthetists and patients in terms of configurative authority. The parametric design tool evaluated by Li and Aflatoony allows OTs to modify existing 3D handgrip models by entering individual hand measurements and selecting from configurable parameters including dimensions, weight, and material [50, 50], with the platform designed so that OTs—not the patients they serve, and not engineers—are the primary operators [50]. This positions OTs as first-line customisers in the assistive technology context, a configuration that could be transferable to prosthetics settings. Similarly, InHwa Lee et al. identify occupational therapists as the source of user needs specifications for the MyoTrainXR system, positioning them as a form of surrogate end-user representative in the design process [42].

Multi-stakeholder platforms that attempt to involve diverse roles simultaneously are described in several sources but tend to resolve toward asymmetric hierarchies when actual configuration decisions must be made. The participatory design protocol of Chu et al. involves an occupational therapist, an engineer, an experienced child user, and a parent in a three-corner collaborative design process [70, 70], but specific mechanisms for mediating between these roles when preferences conflict are not elaborated. The Whatley et al. and Zhou et al. generative co-design systems for aesthetic prosthetics represent a more genuinely symmetric collaboration between algorithm and human user, in which disabled dance artists generate aesthetic seeds through movement that are directly applied to prosthetic covers [29, 35]; here, the technical professional is largely absent from the operational loop, and the platform functions as a relatively neutral aesthetic translation engine. However, these systems address only cosmetic customisation and have not been evaluated in functional prosthetic fitting contexts.

Platform Mechanisms for Design Mediation

Parametric Configuration and Design Automation

Parametric configuration—the use of structured, adjustable parameters to drive design variation—is present across nearly all platform types in the corpus, though with very different levels of user exposure and interpretability. In socket design systems, parametric logic operates largely beneath the user interface, encoding clinical knowledge as automated rules. The SMA² system uses a library of templates and embedded rules drawn from orthopaedic technician practice, generating trim-line positions and rectification operations automatically based on patient measurements [2]. The evidence-generated socket systems of Mbithi et al. extend this logic further, using a statistical shape model and Bayesian inference over 163 previous transtibial CASD records to generate an individualised starting-point socket without clinician input [4, 5]. The clinical trial found no statistically significant difference in socket comfort score between evidence-generated sockets (median 8.6) and clinician-led CAD/CAM controls (median 8.8), with $p = 0.43$ and effect size = 0.05 [5], suggesting that automated parametric generation can match clinical practice at the aggregate level while reducing baseline design effort.

The PACT algorithm for transradial socket design represents the most fully automated parametric design system in the corpus for upper-limb contexts [43]. PACT retrieves the most similar limb-socket pair from a reference library and applies isotropic and anisotropic scaling to match the input limb geometry, delivering a first-draft socket in approximately 13 seconds with surface discrepancies of 2.11 ± 0.51 mm and volume differences of $2.74 \pm 2.56\%$ relative to prosthetist-designed references [43]. Slice-wise area differences were within $\pm 10\%$ for most of the socket length, with larger errors near the proximal trimline and distal tip [43]. The system systematically undersizes sockets for paediatric clients and oversizes them for adults, an age-related bias that the authors identify as a target for rule-based or measurement-informed refinement [43]. This finding illustrates both the power and the limitation of purely data-driven parametric design: the algorithm learns from the distribution of past clinical decisions, including any systematic biases in that distribution.

In control customisation contexts, parametric configuration is more directly visible to both clinicians and users. The CTI of Reznick et al. provides prosthetists with clinically interpretable tuning parameters that are translated internally into model adjustments across multiple walking tasks and sit-stand transitions [68, 68], enabling full multi-activity personalisation with an average iteration cycle of approximately 2 minutes for walking and 1 minute for sit-stand [68]. The Alili et al. user-controlled interface uses a graphical representation of a four-dimensional kinematic preference space, allowing users to modify control points that are then mapped to 12 impedance parameters via reinforcement learning [7, 7]. Average tuning time in that system was 2.5 minutes [7]. These clinician- and user-facing parametric interfaces represent a middle ground between fully automated design and the traditional trial-and-error expert adjustment described in earlier work such as Wells et al. [67].

The Colombo et al. series consistently used low-cost hand-tracking devices—specifically the Leap Motion controller—to make parametric socket manipulation more natural and accessible to prosthetists. Preliminary tests with ten volunteers showed that beginners required repeated gesture attempts while experts performed tasks with fewer trials [33], with the icon-based feedback system found more useful by novices than experts [33]. The progression from SMA in 2010 [21] through SMA² in 2018 [2] reflects an iterative design trajectory in which automation, NUI, and embedded knowledge management were progressively integrated into a unified parametric design environment.

Visualisation and Simulation as Mediation Tools

Visualisation capabilities serve multiple functions in the corpus: as tools for quality assurance, as communication bridges between professionals and patients, and as aids for building intelligibility of platform-generated outputs. The most technically sophisticated visualisation infrastructure in the socket design domain involves finite element

analysis (FEA) coupled with 3D models of the residual limb. In the Morotti virtual testing laboratory, FEA is integrated directly into SMA through automated pipelines that generate pressure distribution colour maps, highlight areas requiring modification, and re-run simulation after each design iteration [12, 12]. The Lee et al. digital design framework for transtibial interfaces similarly uses FEA to simulate limb deformation under load and to design custom liners and sockets, with results supporting statistically significant reductions in pain during walking ($p < 0.05$) and significant decreases in anterior distal tibial pressure ($p < 0.05$) compared with conventional methods [26].

In control and training systems, visualisation functions primarily as a feedback mechanism to close the loop between user action and system state. The Reviewer interface developed by Yang et al. projects EMG signals into a three-dimensional decision space, displaying coloured clusters for each gesture class and a real-time cursor indicating the user's current muscle activity relative to classification boundaries [10]. In a ten-session longitudinal study with 12 able-bodied participants, the Reviewer group showed significantly higher Fitts' Law task completion rates, reduced overshoot, and improved path efficiency compared with a conventional virtual arm visualisation group [10]. The mechanism is meaningful: by making the classifier's internal logic directly visible to the user, the Reviewer supports a form of mutual adaptation in which users can deliberately reshape their EMG patterns to improve separability, rather than relying on trial-and-error adjustment. The Sungelee et al. study on teaching strategies provides complementary evidence, finding that guiding users to train on the least separable gesture class (learner-led strategy) produced larger accuracy gains and more accurate mental models of system behaviour than either random cuing or user-selected training [11]; participants who focused on class separability developed better predictive understanding of prosthesis recognition failures [11].

VR and AR environments introduce a qualitatively different form of visualisation, one that immerses users in a simulated operational context rather than displaying abstract signals. The Górski et al. evaluation of XR technologies in prosthetic development found that Mixed Reality was ranked highest by an expert panel for ease of use, comfort, and fluency, while VR excelled in content quality and realism, and AR received comparatively lower scores despite positive evaluations overall [24, 24]. Notably, older participants with less digital experience reported more difficulties with user interface elements and greater susceptibility to cyber sickness [24], indicating that visualisation modality interacts with user characteristics in ways that affect platform accessibility. The MyoTrainXR commercial VR intervention for myoelectric training achieved regulatory registration and demonstrated increased patient motivation and confidence in clinical pilots, but also surfaced setup complexity and sensor ergonomics as implementation barriers [65]. Liu et al. found in a 2026 comparison study that pattern recognition control showed consistent performance between physical device and VR settings, while direct control and continuous control produced lower performance in VR, likely due to reduced haptic feedback and stricter spatial-alignment requirements [76]; this finding has direct implications for how much confidence can be placed in VR-based evaluation of control modes.

The Vitali 2016 Mixed Reality environment VOLAB, which combined Kinect v2 body tracking, Oculus Rift head-mounted display, and Leap Motion hand tracking, represents an early attempt to create an immersive socket design workspace [16]. Preliminary tests confirmed real-time performance and hand interaction capabilities, but residual limb reconstruction quality was identified as a limitation [16]. AR applications for prosthetic fitting have also been explored in the Fahd Ahmed et al. system, which uses MediaPipe-based pose estimation and AR overlay to enable real-time visualisation of customised prosthetics on the patient's body, with pose detection accuracy within ± 1.2 cm of manual measurements and stable rendering at 28+ FPS [71]. The Vlachostergiou Aquaborg project used an AR app to map speculative prosthetic designs onto the human body for user evaluation, bridging generative design and embodied previewing in a co-creative context [55].

Human-in-the-Loop Assistance and Iterative Adjustment

Human-in-the-loop (HITL) assistance mechanisms vary widely across the corpus in terms of which human is in the loop and at what stage. In socket design platforms, the human in the loop is almost exclusively the technical professional. SMA² structures human involvement through template-based initialisation followed by clinician-guided modification using virtual tools that emulate traditional hand operations [2]; the clinician intervenes at defined decision points while automated processes handle geometric reconstruction and preliminary fitting. The evidence-generated socket systems of Mbithi et al. position human intervention later in the process: the algorithm generates a complete first-pass socket, after which the prosthetist applies the brim line and makes modifications at the fitting session based on patient feedback [4, 4]. This architecture explicitly reserves the prosthetist's time for high-skill customisation and patient communication, with the platform handling baseline design [4, 5].

Iterative adjustment in socket design typically involves cycles of simulated fitting, modification, and re-simulation. In the Morotti virtual testing lab, the system highlights pressure areas exceeding maximum thresholds, the prosthetist or the system automatically adjusts geometry, and the FEA is re-executed [12]; multiple such cycles optimise the socket shape before any physical fabrication occurs. The Bonacini and Buldo digital methodology similarly embeds a deformation-based compression step driven by FEM simulation into the design-before-manufacture workflow, with validation via gait analysis [45, 45]. In these iterative models, decision authority is shared between automated optimisation and human oversight, but the interface does not make this sharing explicitly visible to the user; the prosthetist reviews outputs rather than monitoring or adjusting the optimisation logic itself.

Iterative adjustment in control customisation systems is more visible and interactive. The CTI of Reznick et al. supports a two-minute observe-adjust-reprocess cycle, with prosthetists directly modifying kinematic and impedance parameters and immediately observing the resulting behaviour change [68]. The Quintero et al. CCI uses a virtual constraint scheme in which the clinician designs joint kinematic trajectories that directly drive the prosthesis controller, with the trajectory design itself constituting the primary configuration step [47]; this approach enabled a prosthetist to configure a powered knee-ankle prosthesis for a transfemoral user in under ten minutes, compared with hours of engineer-led parameter search [47]. The Alili et al. user-controlled interface similarly supports rapid iteration, with users selecting kinematic profiles, observing prosthesis behaviour, and revising preferences within a single session [7].

In training systems, iterative adjustment takes the form of progressive learning with feedback. The Perry et al. VR training platform showed significant improvement in active motor scores over time for Basic ($\beta = 0.125$, $p = 0.022$) and Advanced ($\beta = 0.45$, $p = 0.001$) motion sets, with 95% classification accuracy achieved for Basic sets by all participants and for Advanced and Digit sets by 50% of participants [34]. The InHwa Lee et al. MyoTrainXR system showed improvement from $86.1 \pm 5.8\%$ to $98.5 \pm 3.7\%$ success rate for intact-limb participants and from 0–40% to 100% for upper-limb-loss participants across four 45-minute training sessions [42], with median iteration completion time decreasing from 19.4 to 15.8 seconds ($p < 0.05$) [42]. The closed-loop data acquisition strategy of Brusamento et al. improved myocontrol performance by guiding users through acoustic feedback to provide training data in underrepresented regions of the input space [63, 63], outperforming standard open-loop acquisition in a study involving one prosthetic hand and activities inspired by daily living [63].

Variant comparison—the ability to evaluate multiple design options simultaneously or sequentially—is less systematically supported. In socket design, the Gabbiadini framework permits rapid generation of alternative configurations [1], and the SMA² system supports multi-material 3D printing as a means of realising design variants [2], but structured side-by-side comparison tools are not described. In control systems, the Alili et al. blinded comparative study showed that users could reliably identify their preferred control profile when presented with alternatives [7], and Vaynee Sungeelee et al. used a post-training questionnaire to assess participants' predictive understanding of the system's recognition behaviour [11]. Yunmei Liu et al. explicitly evaluated three control modes (direct control, pat-

tern recognition, continuous control) in parallel across VR and physical device settings [76], providing one of the more systematic variant comparisons in the corpus; their finding that pattern recognition was uniquely robust to VR's sensory limitations while direct control showed lower cognitive workload in VR [76] illustrates how structured variant comparison can yield non-obvious design insights.

Supervision and Decision-Making Responsibility

Supervision mechanisms—tools that enable oversight of design decisions by a responsible professional—are most developed in the lower-limb socket design domain, where clinical accountability is clearest. In the SMA² system, experienced technicians can use the platform to mentor junior designers, with the platform's embedded rules providing a form of algorithmic supervision of novice decisions [2, 2]. The Colombo et al. 2011 framework validation involved the technical staff of a certified orthopaedic laboratory reviewing platform-generated configurations against their own expertise [23]. The FEA-based systems of Morotti [12] and Bonacini and Buldo [45] use pressure distribution visualisation as a supervisory feedback mechanism, allowing the prosthetist to identify and correct suboptimal designs before fabrication.

In control customisation systems, supervision is less formalised but still present. The Wells et al. MyoWizard and MyoAssistant system provides graphical display of signals during parameter adjustment, enabling prosthetists to observe the effect of changes before committing them to the device [67]. The Xiaolu Bai et al. framework for intelligent lower-limb prosthesis interaction distinguishes two classes of user-system interaction—direct interaction during locomotion and indirect interaction through a UCI to address current or foreseeable challenges—and notes that certain parameters are accessible only to prosthetists, creating a tiered supervision structure in which clinical oversight of high-stakes settings is preserved [13, 13]. The framework explicitly identifies transparency—the degree to which wearers can understand the LLP's current state and planned actions—as crucial for building trust [13].

Trust and perceived control emerge as recurring themes across user experience findings, particularly in control customisation contexts. Resnik et al.'s qualitative study of the DEKA Arm with EMG-PR control found that the majority of participants preferred their personal prosthetic controls over the EMG-PR system, citing complexity, calibration burden, and fatigue as primary concerns [18]; most were nonetheless optimistic about the future potential of EMG-PR as a method [18]. This pattern—scepticism toward current implementation alongside openness to the underlying technology—recurs in several studies. The Mbithi et al. qualitative interviews found that while EG sockets were statistically non-inferior in comfort, participants valued expert prosthetist input and saw co-design with the clinician as meaningful in itself [4, 5]; the platform could not substitute for the relational and communicative functions of clinical interaction. The Bai et al. review of intelligent LLP interaction similarly identifies the wearer's need to understand the device's goals and behaviour as a prerequisite for trust, and notes that prior knowledge of the UCI significantly affects the wearer's ability to interact with it productively [13].

The intelligibility of automated recommendations is addressed most directly in the Alili et al. study, which found that users required additional explanation of knee kinematics to operate the interface effectively [7], and in the Sungeelee et al. study, where the learner-led training strategy improved users' mental models of the classifier's recognition behaviour [11]. The Reviewer system of Yang et al. was specifically designed to address intelligibility by making the classifier's decision space directly visible, and demonstrated that structured feedback reduces reliance on heuristic recalibration [10]. These findings collectively suggest that intelligibility of platform logic—rather than merely usability of the interface—is a distinct and important dimension of user experience in control customisation systems.

In physical design contexts, intelligibility concerns arise differently. Eshraghi et al. found that clinicians performing digital rectification reported challenges due to the absence of tactile feedback, which in traditional casting allows

prosthetists to detect tissue compliance and apply clinical judgment in a fundamentally embodied way [6]; this represents a form of intelligibility deficit created by digitisation itself, in which the platform's abstractions are less interpretable to experts than the physical material it replaces. Olsen et al. similarly found that the absence of clinical experience in a research-led socket fabrication process led to specific quality failures, and recommended that algorithmic tools such as FEA be used to partially compensate for missing tacit knowledge [14].

Thematic Analysis

Theme 1: The Platform as Structured Professional Tool vs. Neutral Interface

The evidence across the corpus converges on a clear finding: digital prosthetic customisation platforms are not neutral interfaces but structured collaboration spaces that encode existing professional hierarchies and workflows into their architecture. The SMA and SMA² systems were explicitly designed to replicate the operations of orthopaedic technicians using virtual tools [1, 2], making the technician's knowledge and practice the organising principle of the interface. The knowledge-based framework of Gabbiadini embeds expert knowledge via virtual assistants that provide structured workflows and automated assistance, with no direct evidence of end-user involvement in the design process [1]. The clinician control interfaces of Wells et al. [67], Quintero et al. [47], and Reznick et al. [68] similarly position the clinical professional as the authoritative configurator, with the platform functioning as a translator of clinical judgment into device behaviour.

This professional-centric structuring is not uniformly problematic. Several studies demonstrate that embedding clinical knowledge into platform architecture improves design quality and reproducibility. The Lee et al. physics-based digital framework achieved statistically significant improvements in comfort and pain reduction relative to conventional artisanal methods [26], suggesting that algorithmic encoding of biomechanical principles can outperform unaided clinical judgment in specific measurable dimensions. The PACT algorithm's ability to deliver a clinically-tolerant first-draft socket in 13 seconds [43] illustrates the efficiency gains possible when structured professional knowledge is successfully formalised and automated. However, both systems also identify residual roles for clinical expertise: PACT's authors recommend incorporating metadata such as tissue stiffness, age, and clinician-led measurements to improve personalisation [43], while Lee et al. note that manual modifications remain part of the workflow.

Where the professional-centric model encounters its limits is in contexts where end-user needs, values, or preferences cannot be adequately proxied by professional knowledge. The Mbithi et al. qualitative data capture participants expressing the value of expert prosthetist input in terms that go beyond technical quality—they describe the fitting session itself, and the clinician's communicative engagement, as a component of their experience [5]. The Whatley et al. and Zhou et al. aesthetic co-design systems reveal a dimension of prosthetic customisation—identity, expressiveness, agency over bodily appearance—that professional-centric design platforms systematically under-serve [29, 35]. The disabled dancers in those studies reported a strong sense of ownership over designs derived from their movements [29], a form of perceived control unavailable in conventional clinical design workflows.

Theme 2: Automation and the Redistribution of Expertise

A second theme running through the corpus concerns the way increasing automation redistributes expertise rather than simply replacing it. The SMA series illustrates this: initial versions required high levels of tacit knowledge from prosthetists; successive versions incorporated more automated steps (MRI reconstruction, automatic rectification, FEA-based optimisation), while new forms of expertise—in operating the digital interface, interpreting simulation outputs, and using hand-tracking devices—were required in return [2, 12, 16]. This is consistent with the Eshraghi

et al. finding that clinicians performing digital rectification required adaptation to compensate for the absence of tactile feedback [6, 6], and with the Howard et al. finding that non-expert designers require explicit training in digital design and manufacturing to participate in co-design workflows [60].

In control customisation systems, automation acts primarily as a cognitive offload mechanism for end users, reducing the number of degrees of freedom that must be actively managed. The semi-autonomous gaze-EMG interface of Greene et al. demonstrated that when users can command a prosthesis in task-space rather than joint-space, they show significantly reduced compensatory motion of the upper arm ($p \leq 0.05$), more effective utilisation of configuration space ($p \leq 0.01$), and more efficient motion ($p \leq 0.01$) compared with traditional classification-based interfaces [8]. The Cirelli et al. context-aware semiautonomous system, which combines user-initiated grasp selection via EMG with computer vision for wrist orientation and grasp refinement, achieved a 100% success rate (vs. 97.69% for EMG-alone) while reducing task completion time by up to 52.66% and improving system usability scale scores by 15 points [61]. The Stracquadanio et al. eye-in-hand shared-autonomy system similarly enabled faster grasps and simplified user experience relative to multi-DOF manual control [62]. These systems position automation not as a replacement for user agency but as a mechanism for making that agency more effective, an architecture the Chunhao Peng et al. system makes explicit by preserving user authority (via EMG and voice) while allowing autonomous decision-making for speed and efficiency [9].

The Marković sensor-fusion framework provides a theoretical grounding for this redistribution: by equipping the prosthetic hand with computer vision and inertial sensing, the controller takes over contextual adjustment (hand preshaping, orientation) while the user provides only high-level commands via a two-channel myoelectric interface [74, 74]. This architecture simplifies the user's control problem not by reducing the prosthesis's capability but by shifting lower-level decisions to the automated layer. The resulting system is described as both more usable and more capable—an important empirical claim that challenges the assumption that automation necessarily diminishes user control.

However, the redistribution of expertise through automation creates new forms of dependence and new failure modes. Several sources identify the training data dependency as a critical limitation. PACT's age-related biases arise from the distribution of its training library [43]. Evidence-generated socket systems trained on single-centre datasets may not generalise to different clinical populations [5]. The Resnik et al. study found that the specific iteration of EMG-PR used with the DEKA Arm created problems of complexity and calibration that outweighed the system's theoretical benefits for most participants [18], suggesting that automation that requires more user effort to maintain than it saves in operation is unlikely to be adopted. Bai et al.'s framework for intelligent LLP interaction identifies this as a systemic issue: prior knowledge of the UCI affects the wearer's ability to interact productively with the device, and interfaces designed without attention to the learning curve may create barriers rather than reduce them [13, 13].

Theme 3: Co-design as Aspiration and Practice

A third theme concerns the gap between co-design as a stated aspiration and co-design as a practically implemented process. Several sources describe co-design processes in ways that involve genuine bidirectional influence between professionals and users, while others use the language of co-design to describe processes that are primarily consultative.

The most developed co-design implementations in the corpus involve aesthetic and identity-related customisation. The Whatley et al. and Zhou et al. projects embedded disabled dancers directly in the generative algorithm's input loop, with the dancers' movements constituting the primary design data [29, 35]. The Vlachostergiou Aquaborg project iteratively developed speculative wearable designs through co-creative sessions between the researcher-

designer and a single female prosthetic user, explicitly attending to the user's affective and emotional needs alongside physical requirements [55, 55]. The Lin et al. VR collaborative design study used the Collab feature of Gravity Sketch to enable simultaneous modification of virtual prosthetic models by designers, medical technicians, and users [38], with positive assessments from both user participants and prosthesis designers [38].

In clinical and functional customisation contexts, co-design is more typically operationalised as iterative user feedback within a clinician-led design process. The Mbithi et al. studies describe clinicians as responsible for design decisions while patients provide comfort ratings and participate in selection between alternatives [4, 5]. The Howard et al. workshop study identifies barriers to genuine co-design scaling within healthcare services, noting that end-users must be engaged from the outset to set goals and manage expectations, but that current systems lack the digital infrastructure to make feedback easily accessible and actionable across the service ecosystem [60]. The study maps sixteen individual stakeholder categories and ten corporate stakeholder groups as part of the AT co-design ecosystem [60], underscoring the structural complexity of implementing symmetric co-design in clinical environments.

The Peixoto et al. platform demonstrates that co-design is more tractable when it is separated from clinical function and focused on personalisation at the level of aesthetics and identity [40]. By decoupling aesthetic customisation from clinical fitting, the platform creates a domain where users can exercise genuine autonomy without requiring clinical supervision—though the authors acknowledge that the platform's current scope is limited to surface customisation [40].

Theme 4: Access, Equity, and Scalability

A final theme concerns the relationship between platform design and the equitable distribution of prosthetic access. Several sources explicitly frame digital manufacturing and platform-mediated customisation as potential solutions to accessibility inequities. The De Vivo Nicoloso et al. LIMBER workflow for 3D-printed monocoque transtibial prostheses reports reductions in average cost, weight, and production time by 95%, 55%, and 95% respectively compared with traditional devices [77], with the authors identifying rural and under-resourced communities as primary beneficiaries. The Junghun Lee et al. computer-vision-based system for resource-constrained environments uses a mobile phone and open-source software to create virtual limb models and generate customised prosthetic hands with minimal specialist involvement [44, 44], directly targeting the barrier of specialist unavailability. The Romero et al. affordable AI-driven myoelectric prosthesis achieved an average BAM-ULA score of 7.4 out of 10 across nine transradial amputees and demonstrated performance comparable to high-end commercial devices [51], with the authors explicitly addressing the needs of users in low-income regions.

The Oldfrey et al. scoping review of digital fabrication in prosthetics provides a broader perspective on this theme, noting that while digital technologies offer reductions in manual labour and potential for automation, the field is dominated by small, proof-of-concept studies using FDM techniques, with few formal studies on mature technologies, little discussion of training requirements for personnel, and a lack of standardised testing methodologies [15]. The review identifies the absence of long-term, large-scale studies as a critical gap, and calls for greater collaboration between academia, industry, and clinical teams to move from proof-of-concept to evidence-based implementation [15].

Synthesis

The Apparent Contradiction: Automation vs. Retained Clinical Authority

The corpus presents a surface-level contradiction. On one hand, multiple studies demonstrate that automated platform-generated designs are statistically comparable to clinician-generated designs in comfort and fit outcomes (Mbithi et al. EG sockets, Lee et al. FEA interfaces, Pendse PACT algorithm), implying that clinical expertise may be partially substitutable by algorithmic logic. On the other hand, qualitative findings from Mbithi et al. [4, 5], Eshraghi et al. [6], and Howard et al. [60] consistently show that clinical expertise is not merely a technical resource but also a relational, communicative, and contextually adaptive function that platforms have not yet reproduced.

These two findings are reconcilable. The quantitative non-inferiority results are aggregate findings: across a population of users, algorithm-generated sockets perform comparably to clinician-led designs on average. But average performance conceals the distribution of individual outcomes. The EG socket studies report lower variability in comfort scores [4, 5]—a finding that cuts both ways. Reduced variability means fewer poor outcomes, but it may also mean fewer optimised outcomes for individuals with unusual anatomy or complex clinical needs. The PACT algorithm's age-related bias toward undersizing paediatric and oversizing adult sockets [43] is precisely the kind of systematic error that aggregate non-inferiority can mask. Clinical expertise may be most valuable not at the average but at the tails of the distribution, where patients differ most markedly from the training data underlying algorithmic systems.

This framing resolves the contradiction: automation effectively handles baseline design for the modal patient, while clinical expertise remains essential for atypical cases, relational communication, and the integration of contextual information (tissue health, psychological state, activity goals) that algorithms do not currently incorporate. The appropriate architecture is therefore not automation replacing the clinician but automation restructuring the clinician's work, moving their contribution from routine design to high-value customisation and patient collaboration—which is precisely the design intention articulated in the Mbithi et al. series [4, 5] and the PACT algorithm [43].

Why Control Interface Studies and Socket Design Studies Diverge

A second tension in the corpus concerns the different patterns of end-user participation found in socket design versus control customisation research. End users are largely absent as configurative agents in socket design platforms but are progressively central in control customisation systems. This divergence reflects genuine differences in the nature of the configuration task rather than differing philosophies of user involvement.

Socket geometry is determined by tissue biomechanics, load transfer requirements, and residual limb anatomy—domains in which the patient lacks the technical knowledge to make informed parametric adjustments without clinical mediation. Control preference, by contrast, is inherently subjective: the user alone knows how a given control mapping feels during task performance. The Alili et al. blinded study provides direct evidence for this, showing that users can reliably identify their preferred control profiles when presented with alternatives [7], while their clinicians cannot make that determination on their behalf. Similarly, Arunachalam et al. propose that control optimisation should be structured around the user's own cost functions (effort, time, reliability) rather than a clinician-defined objective [49, 49], and Garcia-Rosas et al. demonstrate that implicit human motor behaviour—specifically upper-body compensation, measurable via wearable IMUs—can serve as an objective function for automatic prosthesis personalisation without requiring explicit user feedback [58, 58].

These findings collectively suggest that the appropriate boundary between clinical authority and user authority differs by design domain. In socket geometry, clinical authority should dominate with user input as a comfort check. In control preference, user authority should dominate with clinical oversight ensuring safety and technical feasibility.

Current platforms that apply a single collaboration model across both domains are likely to either over-involve users in technical decisions they cannot meaningfully evaluate or under-involve them in preference decisions that only they can make.

The Evidence Gap for Upper-Limb Socket Design

The most significant structural gap in the corpus concerns upper-limb socket design specifically. While there is substantial evidence from lower-limb socket design platforms (the Colombo/Vitali/Gabbiadini/Morotti series, the Mbithi evidence-generated socket studies, the Lee FEA framework, the Bonacini and Buldo methodology), analogous digital platforms for trans-radial, trans-humeral, or more proximal upper-limb sockets are represented by only a few sources: the Olsen et al. study of a research-led trans-radial socket fabrication process [14], the Eshraghi et al. feasibility study of digital vs. manual socket design for below-knee amputees (relevant by analogy) [6], the Pendse PACT algorithm for transradial sockets [43], and the Binedell et al. forequarter prosthesis case study [20]. Olsen et al. explicitly note that modernising upper-limb prosthetic socket manufacturing has seen limited progress, with casting techniques resembling those developed over 50 years ago [14], and the Oldfrey et al. scoping review identifies a lack of long-term large-scale studies as a critical limitation of the current digital fabrication evidence base [15].

This gap is consequential for the research question. Upper-limb socket fit has different biomechanical constraints from lower-limb fit, with different pressure tolerances, different functional demands during manipulation tasks, and different relationships between socket geometry and prosthetic control signal quality. Whether the design paradigms, parametric tools, FEA models, and evidence-generation methods developed for lower-limb sockets transfer to upper-limb contexts is an empirical question that the literature has not yet adequately addressed. The PACT algorithm's results for transradial sockets [43] represent a beginning, but with $n = 19$ and identified age-related biases, it does not yet constitute mature evidence. Future work that directly resolves this gap would require prospective clinical studies comparing digital and traditional socket design approaches for upper-limb amputees using comfort, control quality, and functional performance as joint outcomes—a study design not currently represented in the corpus.

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